ON THE CHOICE OF TURBULENCE MODEL FOR THE SIMULATION OF AIRFOILS AT REYNOLDS NUMBER BELOW 200,000

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Abstract

Computational fluid dynamics is increasingly used in the design and optimisation of wings of small unmanned aerial vehicles. Airfoils of these small vehicles operate at Reynolds numbers where laminar-turbulent transition has a pronounced effect on performance. Selecting the most adequate transitional turbulence model is thus critical. Despite this, a thorough validation of various transition models at Reynolds numbers below 200,000 has not been conducted to date. This paper presents a validation of four transition models ($\gamma - Re_\theta$, $k_T - k_L - \omega$, low Reynolds $k-\omega$ SST, Spalart-Allmaras with transition) using a steady RANS solver for three different airfoils at Reynolds numbers between 60,000 and 200,000. Results show that the $\gamma - Re_\theta$ transition model gives the best prediction of the locations of laminar separation, transition and turbulent reattachment, as well as the aerodynamic performance at low to medium lift coefficients. The low Reynolds $k-\omega$ SST model is more accurate near the top corner of the low-drag region. The Spalart-Allmaras with transition is accurate at a Reynolds number of 200,000, whereas the $k_T - k_L - \omega$ model is not capable of capturing the aerodynamic performance at the conditions tested.

Keywords: Airfoil analysis; Computational Fluid Dynamics; Low Reynolds number; Transition

1. Introduction

Small fixed-wing Unmanned Aerial Vehicles (UAVs) are increasingly used for a variety of reconnaissance and remote-sensing missions, aerial photography, environmental monitoring, disaster relief, and search and rescue operations [1–11]. Airfoils of these small UAVs operate at Reynolds numbers (Re) between 50,000 and 250,000 [12]. These Reynolds numbers are low enough for a significant laminar boundary layer to be present. This low-energy laminar boundary layer easily detaches and can transition and subsequently reattach, forming a so-called laminar separation bubble [13–29]. Bubbles profoundly affect airfoil aerodynamics; they may create non-linearities in the lift curve [30–32], and they increase the section's pressure drag [33]. Bubble size and transition location furthermore depend on the airfoil geometry, angle of attack, Reynolds number, and turbulence intensity [34].

Low Reynolds number aerodynamics has, due to its complex nature, seen a sustained research effort that spans multiple decades [30–33, 35–40]. While the bulk of this work focuses on analysis, testing, and inverse design, more recently deterministic and robust optimisation of low Reynolds airfoils has gained traction [41–44]. This optimisation hinges critically on the ability to predict transition from laminar to turbulent flow. Transition prediction is, however, still immature, both at low Reynolds numbers and for natural laminar flow airfoils at much higher Reynolds numbers than investigated here [45–47].

Several studies have validated how accurately airfoil aerodynamic performance can be predicted [47–54]. Yet, most of these studies focus on the higher end of the Reynolds number range of interest for small unmanned aerial vehicles. Those studies that do investigate low Reynolds number cases, are typically restricted to Reynolds numbers above 200,000 [47, 49–54]. To the best of the authors' knowledge a thorough investigation around the so-called critical Reynolds number, where smooth airfoil performance degrades quickly [35], has not been conducted to date. The main contribution of this work is thus to investigate if the transition models available in open-source Computational Fluid Dynamics (CFD) codes can accurately predict the aerodynamic performance and the transition locations of airfoils at Reynolds numbers below 200,000. The investigation presented here additionally covers a significantly larger set of airfoils than any prior low Reynolds number validation study, and provides a critical assessment of transition modelling capability in two of the most widely used open-source CFD codes: OpenFOAM and SU2. Previous work primarily uses either commercial CFD codes [48–50], or codes with restricted access [47, 51–53]. Results presented here are limited to steady Reynolds Averaged Navier-Stokes (RANS) to ensure that computational cost remains moderate so that integration in optimisation studies is possible.

The remainder of this paper is organised as follows. The theoretical formulation section describes transition modeling and the transition models used in this work. The numerical procedure section explains the mesh generator, boundary conditions, and numerical schemes used for the CFD simulations. In the results and discussion section, the predicted aerodynamic performance of three airfoils (E387, FX63-137, and NACA0018) ranging from 9.1% to 18% thickness are investigated. The predicted laminar separation bubble locations of the three airfoils are also compared to experimental results and the findings are discussed. The conclusion section then highlights the main outcomes of the study.

2. Theoretical Formulation

This section will briefly outline the transition models used in this study. More details on each transition model can be found in the reference articles [55–58]. All CFD simulations used within this article are based on the steady Reynolds Averaged Navier-Stokes (RANS) equations. RANS simulations are prominent in the CFD community due to their relatively low computational cost and comparatively accurate results [34]. Only transition turbulence closure models are used in this study to ensure that the low Reynolds number flow phenomena is captured. These transition models build on general one and two equation turbulence models. The different types of transition models will be discussed, followed by the CFD models used within this study.

2.1 Transition Modeling

The transition process that is associated with laminar separation bubbles is complicated and made up of different coherent structures [29, 34]. These coherent structures are unsteady and generally require a Direct Numerical Simulation (DNS), Large Eddy Simulation (LES), or Detached Eddy Simulation (DES) to predict relevant flow features accurately [34]. These simulations, however, are very computationally expensive, with LES being upwards of 1000 times more expensive than an equivalent steady RANS simulation [59]. Due to the averaging process within a steady RANS simulation, details of the transition process are essentially lost and therefore transition can only be obtained through some form of artificial triggering [34, 56]. RANS transition modelling can be obtained in one of three ways: damping functions, empirical relations or a phenomenological framework [34, 58].

The damping function method does not have transition built into the model but limits the amount of turbulence produced in the viscous sublayer until a critical Reynolds number is reached, and the flow is then forced to transition [34, 60]. The low Reynolds $k-\omega$ SST model falls into this category, where the damping function is optimised for an incompressible flat plate boundary layer [56].

Correlation based methods do not attempt to model the physics of transition, but use experimental correlations to trigger transition [34, 60]. Transition is either modelled as instantaneous or with some form of zone length usually based on an intermittency profile [58]. Empirical methods generally predict transition based on the momentum thickness Reynolds number at different streamwise locations. When the momentum thickness Reynolds number exceeds an empirically defined value, transition is initiated [34]. The $\gamma-Re_\theta$ model uses the relationship between the local vorticity Reynolds number and the momentum thickness Reynolds number, with the latter used to trigger transition [60]. The

Bas-Cakmakcioglu Spalart-Allmaras algebraic transition model also uses the local vorticity Reynolds number to predict the onset of transition [55].

Physics-based transition models attempt to model the transition related phenomena through functions of dimensionless quantities known as sensors [34, 58, 61]. Some of these models introduce the concept of laminar kinetic energy to represent the pre-transitional fluctuations in laminar attached or separated boundary layers, whereas some other models use the energy of fluctuating velocity [58, 61]. The $k_T - k_L - \omega$ model uses the ratio between the turbulent production time-scale and the molecular diffusion time-scale to initiate transition [58].

2.2 Low Reynolds correction for $k-\omega$ SST two-equation model

The low Reynolds variant of the standard $k-\omega$ Shear Stress Transport (SST) is a modification to the baseline $k-\omega$ SST model which uses a damping function to initiate transition (details can be found in ref. [56]). Artificial damping of turbulence occurs until the local Reynolds number reaches a critical value. At this point the damping function causes a significant growth in turbulent kinetic energy and in the specific turbulence dissipation rate [34, 56]. Further downstream, the production and dissipation of turbulent kinetic energy balance each other, which indicates a fully turbulent flow [34]. The critical local Reynolds number of the low Reynolds $k-\omega$ SST model is optimised for an incompressible flat-plate boundary layer and therefore variables within the model need to be tailored to specific cases [56]. These variables were not tailored within this study as the objective was to see the predictive capability of the model as it was originally designed.

2.3 Langtry-Menter four-equation transitional SST model ($\gamma - Re_\theta$)

The correlation-based transitional modification of the $k-\omega$ SST model is called the Langtry-Menter four-equation model or the $\gamma - Re_\theta$ model (details can be found in ref. [57, 60]). The original $k-\omega$ SST model is coupled with two additional equations, one for intermittency γ, and one for the transition onset momentum thickness Reynolds number Re_θ. The $\gamma - Re_\theta$ model captures variations in turbulence intensity, as well as the influence of the freestream and the pressure gradient through the transition momentum thickness Reynolds number transport equation [34, 60]. The transition Reynolds number is initiated in the freestream through experimental correlations and then diffused into the boundary layer [34]. The critical momentum thickness Reynolds number ($Re_{\theta c}$), where the intermittency is first triggered, is correlated empirically with the local transition momentum thickness Reynolds number ($\widetilde{Re}_{\theta t}$) [60]:

$$Re_{\theta c} = f(\widetilde{Re}_{\theta t}) \tag{1}$$

based on a series of experiments on flat plates [60]. Once the critical value of $Re_{\theta c}$ is met, intermittency production is triggered and an empirical correlation is used to determine the length of the transition zone (F_{length}) [57]:

$$F_{length} = f(\widetilde{Re}_{\theta t}) \tag{2}$$

When the intermittency function is triggered, the turbulent kinetic energy production term is switched on, which causes the flow to transition from laminar to turbulent. The physics of transition for the $\gamma - Re_\theta$ model is contained entirely in the experimental correlations [60].

2.4 Spalart-Allmaras one-equation Bas-Cakmakcioglu transitional model

Cakmakcioglu [55] propose a version of the SA turbulence model[1] that can predict transition. It will be denoted here as the SA-BC transitional model. The SA-BC model predicts transitional flow behaviour by modifying the original SA transport equation. As stated earlier, the SA-BC transitional model is an empirical transition model that compares the local momentum thickness Reynolds number (calculated from the local vorticity Reynolds number (Re_v)) to an experimentally correlated critical

[1] Details of the baseline model can be found in ref [62].

momentum thickness Reynolds number ($Re_{\theta c}$) to trigger the onset of transition [55]. Transition is accomplished by multiplying the turbulence production term in the SA transport equation with an intermittency distribution function which goes from γ_{BC}=0 (laminar) to γ_{BC}=1 (turbulent) [55]. The intermittency distribution function is given by [55]:

$$\gamma_{BC} = 1 - \exp\left(-\sqrt{Term1} - \sqrt{Term2}\right) \tag{3}$$

where Term1 is a function of Re_v and $Re_{\theta c}$ and Term2 is a function of the turbulent viscosity, the local velocity, and the distance to the nearest wall [55]. This intermittency distribution function damps the turbulence production term in laminar flow regions until the transition criterion (local vorticity Reynolds number) is met. At this point the damping effect is disabled and the flow is allowed to be fully turbulent [55]. The SA-BC transition model has no transition length correlation, and is calibrated for a zero pressure gradient flat plate case with a turbulence intensity of 0.2% and a Reynolds number of 3.6 million [55].

2.5 $k_T - k_L - \omega$ three-equation model

The $k_T - k_L - \omega$ transition model is a three-equation model proposed by Walters and Cokljat [58] that is based on the $k - \omega$ model with an additional laminar kinetic energy transport equation. This model is a physics based model that attempts to capture the low frequency velocity fluctuations in the pre-transitional region, as this has been shown to be an antecedent to transition [58]. Walters and Cokljat [58] state that these pre-transitional fluctuations include the Klebanoff modes and Tollmien-Schlichting waves that are precursors to bypass and natural transition, respectively. The $k_T - k_L - \omega$ model initiates transition when the ratio between the turbulent production time-scale and the molecular diffusion time-scale reaches a critical value, and this critical value is calculated through threshold functions for both bypass and natural transition [58]. The threshold functions (β) for bypass (BP) and natural (NAT) transition are [58]:

$$\beta_{BP} = 1 - \exp\left(-\frac{\phi_{BP}}{A_{BP}}\right) \tag{4}$$

$$\beta_{NAT} = 1 - \exp\left(-\frac{\phi_{NAT}}{A_{NAT}}\right) \tag{5}$$

where A is a model constant and ϕ is a function of the turbulent kinetic energy for bypass transition and the laminar kinetic energy for natural transition [58]. These threshold functions are part of the model terms that appear in the three transport equations. Transition occurs through a transfer of energy from laminar kinetic energy to turbulent kinetic energy [58].

2.6 Laminar Separation Bubble Prediction

As mentioned previously, laminar separation bubbles can form in the presence of a strong adverse pressure gradient along the aerodynamic surface. This adverse pressure gradient can cause the laminar boundary layer to separate. Small disturbances present in this separated boundary layer are subsequently amplified in the shear layer causing a transition to turbulent flow to take place [27, 29]. As this turbulent flow carries high momentum normal to the shear layer, it can reattach forming a closed bubble which is known as a laminar separation bubble. The presence of this bubble influences the global aerodynamic coefficients as the bubble displaces the outer inviscid flow [27]. This reduces the strength of the suction near the leading edge and decreases the pressure recovery near the trailing edge which, in turn, increases pressure drag [27].

As the laminar separation bubble has a strong impact on total drag, CFD calculations need to predict the bubble size and location correctly to predict the aerodynamic performance well. The location of laminar separation, transition onset, transition, and turbulent reattachment can be determined from the local skin friction coefficient C_f on the airfoil surface (Figure 1). Laminar separation occurs where the local skin friction coefficient turns negative, while turbulent reattachment takes place when the local skin friction coefficient returns to a positive value [51]. The transition onset is where the skin friction decreases after separation. Transition appears where the skin friction coefficient is at its minimum.

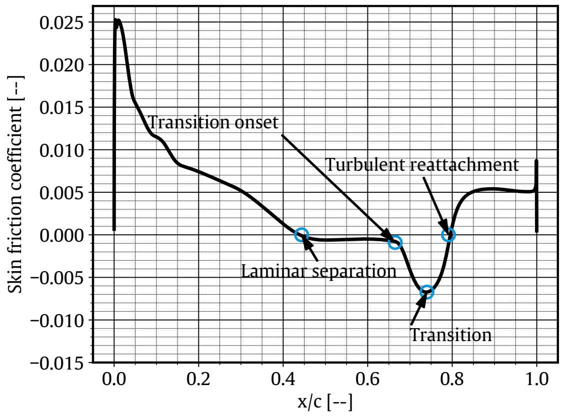


Figure 1 – Upper surface skin friction coefficient of the E387 airfoil at a Reynolds number of 100,000 and 2° angle of attack for the $\gamma - Re_\theta$ model

3. Numerical Procedure and Grid Convergence

This section details how the meshes are generated, and explains the boundary conditions and numerical schemes used in the CFD simulations. A grid convergence study is also presented.

3.1 Mesh Generation and Boundary Conditions

A fully structured O-grid mesh is adopted for all CFD simulations presented here. The airfoil is located in the middle of the mesh and placed 20 chords away from the boundaries. The airfoil surface is defined with 501 points, and a first cell height (y^+) value of 0.5 is used to capture the boundary layer region accurately (the choice of mesh settings will be addressed in the next section). An O-grid mesh topology is selected to avoid the very high aspect ratio cells near the trailing edge of the airfoil typically found with a C-grid. With O-grids the same mesh can also be used for all angles of attack by simply applying appropriate flow direction vector components [63]. To allow an O-grid topology, airfoils with a sharp trailing edge are given a trailing edge thickness of 0.25% of the chord length. The mesh is produced using an open-source mesh generator Construct2D [64]. The mesh for the E387 is shown in Figure 2.

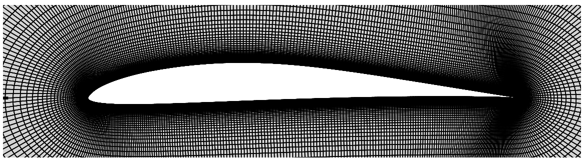


Figure 2 – Mesh generated by Construct2D for the E387 airfoil

Freestream values for all CFD variables are specified on the farfield boundary of the mesh and no-slip wall boundary conditions are applied at the airfoil surface. The fluid inside the domain is defined as air with a density of 1.225 kg/m^3 and a dynamic viscosity of 1.789×10^{-5} Pa.s.

3.2 CFD Simulation Procedure

Simulations presented here are performed with the open-source CFD solvers OpenFOAM and SU2. OpenFOAM is used for the low Reynolds $k - \omega$ SST, $\gamma - Re_\theta$, and $k_T - k_L - \omega$ transition models, while the SA-BC transition model is simulated in SU2. A steady calculation is performed using OpenFOAM's pressure-based solver, SIMPLE (Semi Implicit Method for Pressure Linked Equations), and an incompressible RANS solver is employed for both OpenFOAM and SU2.

3.2.1 OpenFOAM

For OpenFOAM simulations, convective terms are discretised with the second-order linear-upwind discretisation, and a linear discretisation is employed for the Laplacian terms. Using a second-order scheme results in drag predictions that match experimental values much better than a first-order upwind discretisation where drag is over-predicted by close to 50%. The linear equations for kinetic pressure p_k (the ratio of static pressure to density) are solved using Generalised Geometric-Algebraic Multi-Grid (GAMG) solver with Gauss-Seidel as a smoother with a tolerance of 10^{-10}, while the linear equations for the rest of the velocity and turbulence variables are solved using Gauss-Seidel solver with a tolerance of 10^{-8}. A relaxation factor of 0.3 is used for p_k and a relaxation factor of 0.7 is used for the velocity and turbulence variables. A change in tolerance from 10^{-5} and 10^{-7} alters drag coefficients with less than 2%. To reduce computational time, a residual tolerance of 10^{-5} is thus set for all variables.

3.2.2 SU2

Stanford University Unstructured (SU2) is used to simulate airfoils with the SA-BC turbulence model [65]. With SU2, a Green-Gauss cell-based discretisation scheme is adopted for spatial gradients, and an Euler implicit discretisation is chosen for temporal discretisation. The linear equations are solved using Flexible Generalised Minimum Residual (FGMRES) and Incomplete Lower Upper factorisation with connectivity-based sparse pattern (ILU) as the linear pre-conditioners. An upwind scheme, Flux Difference Splitting with low-speed preconditioning (FDS), is used for the convective terms.

SU2 generally requires a smaller tolerance than that used in OpenFOAM. While drag reduces by more than 5% for a reduction in tolerance from 10^{-5} to 10^{-7}, the difference in drag is less than 1% when the residual tolerance is changed from 10^{-7} to 10^{-8}. As SU2 generally converges quickly and requires less computational time then OpenFOAM, a residual tolerance of 10^{-8} is chosen for the residual of the RMS pressure.

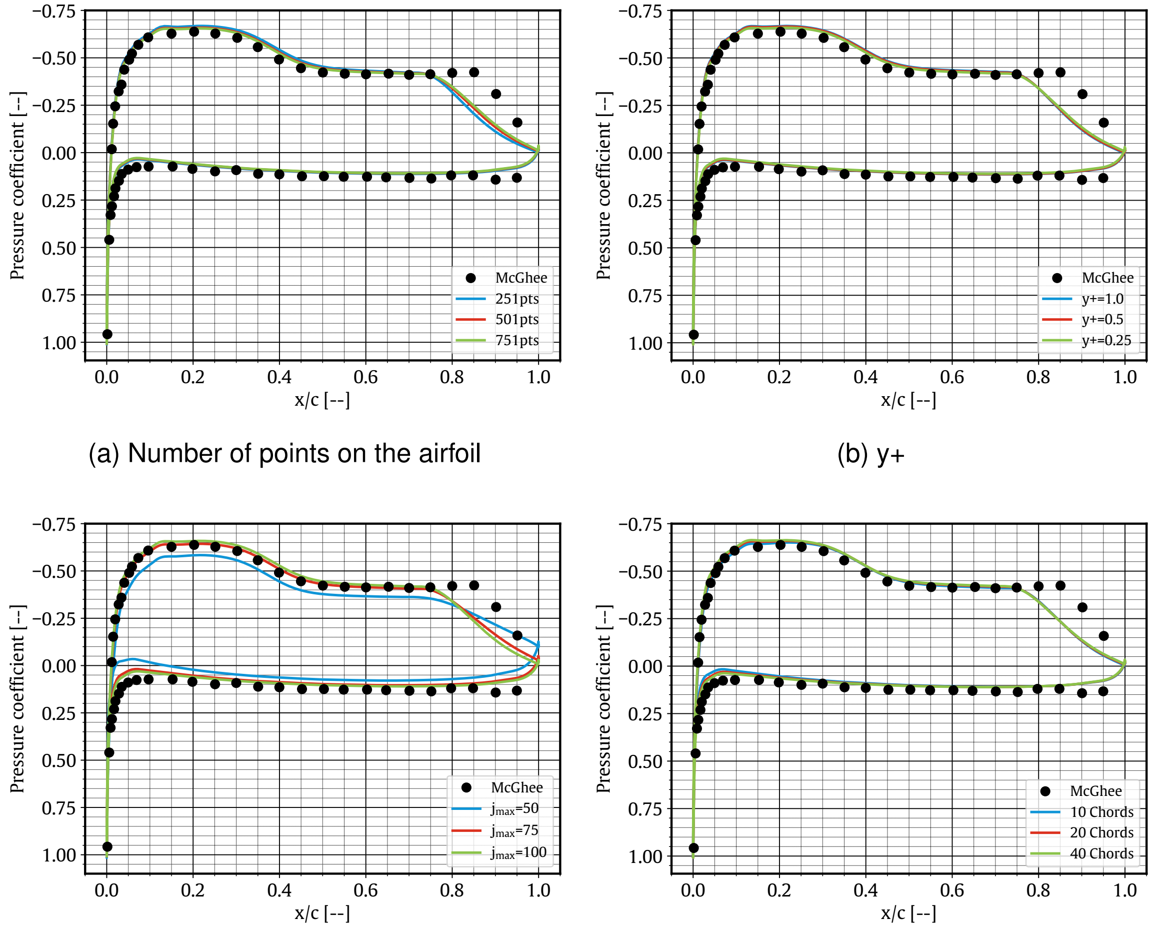
Accurate prediction of transition locations with the SA-BC model, however, requires a convergence tolerance of 10^{-14}. With a standard tolerance of 10^{-8}, the SA-BC transition model fails to predict the transition location accurately across the entire range of angles of attack. This change in tolerance does not influence the drag prediction significantly.

3.3 Grid Convergence

A grid convergence study is performed on the E387 airfoil at a chord Reynolds number of 60,000 and an angle of attack of 2°. Mesh settings such as the number of points on the airfoil's surface (n), the first cell layer height y^+, the number of points from the airfoil surface to the boundaries of the mesh (j_{max}) and the radius of the domain (r) are varied. j_{max} determines how coarse or fine the farfield mesh is which has the largest impact on the computational time. The results for each mesh setting are compared and results for the $\gamma - Re_\theta$ turbulence model are presented as they are representative of the behaviour of all transition models.

The pressure coefficient $C_p = (p - p_\infty) / \left(0.5 \rho V_\infty^2\right)$ is compared for the various meshes. Here p_∞ is the atmospheric pressure and V_∞ is the freestream velocity. Figure 3 presents the effect of mesh settings on the pressure coefficient on the top and the bottom surface of the airfoil. The baseline mesh settings are $n = 501$, $j_{max} = 100$, $y^+ = 0.5$, $r = 20$ chords and only one setting is changed in each of the grid convergence plots while other settings are kept at their baseline values. Overall, the pressure coefficients show good agreement with experimental data apart from the premature transition and reattachment. Transition can be identified by the sharp increase in C_p near the trailing edge of the airfoil. However, small differences can be observed for different mesh settings. As expected, these small differences mainly occur in the transition region.

Figure 3a shows that using 501 and 751 points on the airfoil surface results in a transition location that is slightly aft of that obtained with 251 points. The results with 501 and 751 points are almost indistinguishable which suggests that the results are independent of the number of points on the airfoil when using at least 501 points.



(a) Number of points on the airfoil

(b) y+

(c) Number of points normal to airfoil

(d) Radius of domain

Figure 3 – Effect of mesh settings on the pressure coefficient compared to experimental data [66]

Figure 3b highlights that the C_p distribution on the surface of the airfoil is not largely affected by y^+ for the conditions investigated here. The number of radial points across the domain, j_{max}, on the other hand, does affect the C_p distribution (Figure 3c). A smaller number of radial cells impacts the slope of the increase in C_p across the transition location. With a smaller j_{max}, C_p increases more gradually. Similarly, a variation of the radius of the domain (Figure 3d) slightly changes the slope of the increase in C_p across the transition. The suction peak is similar for all values tested, although a smaller j_{max} does result in a reduced suction peak.

Tables 1, 2, 3, and 4 report the grid sensitivity results for the lift coefficient $C_l = L/\left(0.5\rho V_\infty^2\right)$ and drag coefficient $C_d = D/\left(0.5\rho V_\infty^2\right)$[2]. The differences in the aerodynamic coefficients are all less than 2%, with the exception of the drag coefficient between j_{max} of 50 and 75. A smaller difference in lift and drag coefficients can be observed when j_{max} increases from 75 to 100 suggesting that the results are converging. This is confirmed by the value obtained from the Richardson extrapolation.

The mesh settings used for the remainder of the study are 501 points on the airfoil surface, a y^+ value of 0.5, a j_{max} value of 100, and a domain radius of 20 chord lengths. These values are chosen as a balance between computational expense and simulation accuracy. The same mesh is used for both OpenFOAM and SU2 simulations.

4. Results and Discussion

This section outlines the grid convergence studies, a discussion on the prediction of the laminar separation bubble, and compares the numerical results from the four transition models for the aerodynamic

[2]Area is omitted in the lift and drag coefficient definition because 2D simulations are performed. The lift, L and drag, D obtained from CFD simulations are the lift and drag per unit area.

Table 1 – Grid convergence results for aerodynamic coefficients with change in number of points on the airfoil

n	# Cells	C_l [–]	C_d [–]	ΔC_l [%]	ΔC_d [%]
251	0.5×10^5	0.5286	0.0271	-	-
501	1.0×10^5	0.5230	0.0280	-1.057	3.273
751	1.5×10^5	0.5208	0.0284	-0.420	1.475
∞	∞	0.5194	0.0288	-0.274	1.266

Table 2 – Grid convergence results for aerodynamic coefficients with change in the y^+

y^+	C_l [–]	C_d [–]	ΔC_l [%]	ΔC_d [%]
1.00	0.5536	0.0275	-	-
0.50	0.5296	0.0280	-0.743	1.967
0.25	0.5289	0.0282	-0.131	0.589
0.00	0.3288	0.0282	-0.028	0.319

Table 3 – Grid convergence results for aerodynamic coefficients with change in the number of points normal to airfoil

j_{max}	# Cells	C_l [–]	C_d [–]	ΔC_l [%]	ΔC_d [%]
50	0.50×10^5	0.4347	0.0312	-	-
75	0.75×10^5	0.5070	0.0289	16.62	-7.396
100	1.00×10^5	0.5230	0.0280	3.156	-3.040
∞	∞	0.5274	0.0275	0.870	-1.927

Table 4 – Grid convergence results for aerodynamic coefficients with change in the radius of the domain

r	C_l [–]	C_d [–]	ΔC_l [%]	ΔC_d [%]
10	0.5119	0.0287	-	-
20	0.5230	0.0280	2.168	-2.569
40	0.5255	0.0278	0.477	-0.619
∞	0.5262	0.0278	0.138	-0.191

coefficients and laminar separation bubble locations to published experimental data. A total of three airfoils are investigated to determine how well each transition model predicts the aerodynamic performance (E387, FX63-137 and NACA0018) and the location and size of the laminar separation bubble will be analysed at low Reynolds numbers. Each airfoil will be analysed in its own section. Xfoil (a potential flow panel method with an integral boundary layer formulation [67]) is also examined to see the validity of the method in predicting the location and size of the laminar separation bubble. A turbulence intensity of 0.1% is used throughout the study to match experimental conditions [66, 68–70]. For the NACA0018 a turbulence intensity of 0.02% is used for the aerodynamic performance while a turbulence intensity of 0.19% is used for the prediction of the laminar separation bubble to match experimental conditions [71, 72].

In the proceeding sections, bubble-related locations for the transition models and Xfoil are compared to experimental values for the E387, FX63-137, and NACA0018 airfoils. Xfoil is included as its capability to predict separation, transition, and reattachment locations is not as widely investigated in literature as its predictive capability for aerodynamic performance [50, 73–75]. Xfoil was run with a critical amplification factor n_{crit} of 8.1 to match a turbulence intensity of 0.1%.

4.1 Eppler 387

The Eppler 387 (E387) airfoil is designed to generate moderately high lift coefficients at low Reynolds numbers and is intended for use in model sailplanes [76]. The airfoil (Figure 4) has a maximum thickness of 9.1%, and a maximum camber of 3.8%. The E387 airfoil has been extensively studied computationally and was tested in different wind tunnels [47, 49–51, 66, 68, 77–81]. At the low Reynolds numbers investigated here, considerable differences exist between the values measured in the different wind tunnel campaigns. These likely arise from differences in turbulence intensity of the wind tunnels, variations in model surface finishes and accuracy, measurement techniques, or uncertainties associated with the small forces at low Reynolds numbers [79]. This section investigates the aerodynamic performance and laminar separation bubble predictions of the E387 airfoil for the four transition models, and casts a more extensive look into the coefficient of pressure and flow field results at critical conditions.

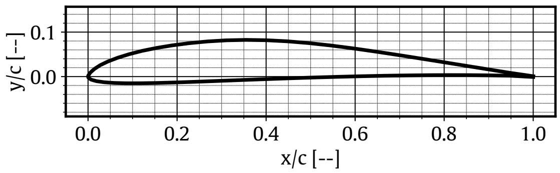
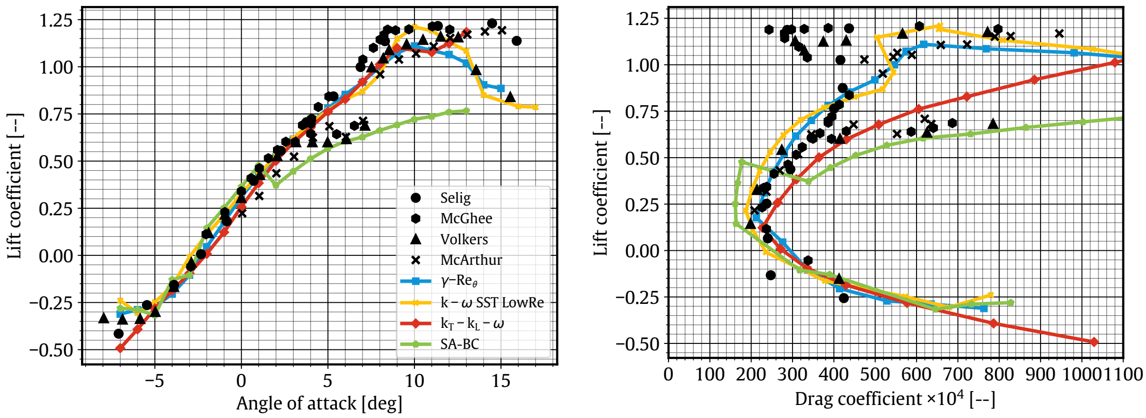


Figure 4 – E387 airfoil

Figures 5 and 6 compare the aerodynamic performance predicted by the four transition models for the E387 airfoil to experimental data (from refs. [66, 68, 78–80]).



(a) Lift coefficient versus angle of attack (b) Lift coefficient versus drag coefficient

Figure 5 – Comparison between the transition models and experimental data of the E387 airfoil at a Reynolds number of 60,000 [66, 68, 78–80]

Figure 5a presents lift coefficients for the E387 airfoil at a Reynolds number of 60,000. As shown, the choice of transition model greatly impacts the predicted maximum lift and the stall angle of the airfoil[3]. The SA-BC transition model predicts lift poorly at an angle of attack greater than 2°, although it is the only model that predicts the decrease in lift around 5° to 7° observed in some of the experiments. SA-BC generally under-predicts lift and the difference between the predicted and experimental lift increases with an increase in angle of attack. The low Reynolds $k-\omega$ SST model, on the other hand, agrees much better with the experimental data, and the maximum lift is predicted well. However, it slightly under-predicts the stall angle. Lift coefficients obtained from the $\gamma-Re_\theta$ model also match experimental data from Volkers [78] and McArthur [79] in the linear region and prior to stall. The maximum lift coefficient is predicted well but the stall angle is slightly under-predicted. Finally, the $k_T-k_L-\omega$ transition model tends to under-predict lift in the linear region and the lift distribution around the stall region is similar to that obtained with the $\gamma-Re_\theta$ model.

While there is mixed agreement with experiments for lift predictions, the results for the various transition models differ even more for drag (Figure 5b). The $k_T-k_L-\omega$ and SA-BC models match the

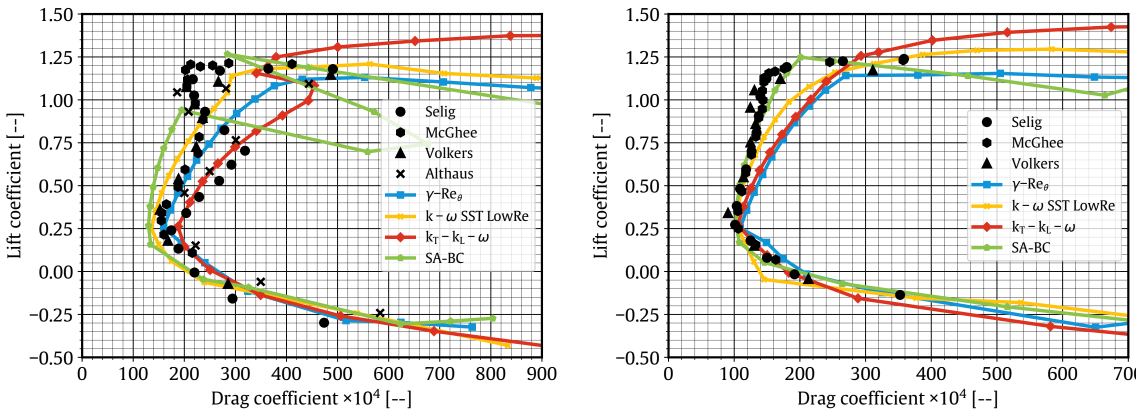
[3]Whereas the overall trend of the aerodynamic coefficients is identical across the different experiments, a difference in maximum lift coefficient and stall angle can be observed.

experimental increase in drag in the region between 4° and 7° at a Reynolds number of 60,000. According to McArthur [79], this drag increase is caused by laminar separation with no reattachment.

At lift coefficients higher than approximately 0.8 a subsequent reduction in drag is measured prior to stall (Figure 5b). At these higher angles the flow reattaches onto the surface of the airfoil after the period with full laminar separation at medium lift coefficients [66, 77, 79]. As the angle of attack is increased, the adverse pressure gradient increases, which causes a rapid forward movement of the transition point [82]. This, in turn, causes reattachment and the formation of a laminar separation bubble that gradually moves towards the leading edge as the angle of attack is increased [66, 79, 83]. None of the investigated transition models are able to fully capture this, possibly because of the unsteady nature of this phenomenon. The low Reynolds $k-\omega$ SST model shows some decrease in drag, although the predicted drag values do not match well with experimental results. However, the large spread in experimental results for this region of the drag polar complicates validation. The $\gamma-Re_\theta$ transition model offers the most accurate drag prediction for the majority of lift coefficients at a Reynolds number of 60,000, with the low Reynolds $k-\omega$ SST model also offering an accurate drag prediction for the majority of lift coefficients.

Figure 6 compares the experimental and numerical drag polars of the E387 airfoil at Reynolds numbers of 100,000 and 200,000. As shown, drag is predicted well around the bottom corner of the low-drag region (drag bucket) for most of the transition models but predictions are less accurate at moderate to high lift coefficients. This suggests that the size of the laminar separation bubble is not well predicted at higher angles of attack [47, 50]. The low Reynolds $k-\omega$ SST model slightly underpredicts drag near the bottom corner of the drag bucket, but performs well near the top corner at a Reynolds number of 100,000. At a Reynolds number of 200,000, its drag prediction is improved, although the drag is over estimated at higher lift values (Figure 6b). The $k_T-k_L-\omega$ transition model predicts drag the closest to experimental data by Selig [68] at a Reynolds number of 100,000 and lower lift coefficients but, once again, the large spread of experimental data further complicates the validation (Figure 6a). The $\gamma-Re_\theta$ model over predicts drag at high lift coefficients for both Reynolds numbers (Figure 6). The drag prediction of the SA-BC transition model at a Reynolds number 200,000 agrees very well with experimental data [66, 68, 78], but the model under predicts drag at a Reynolds number of 100,000 for lift coefficients up to around 1.0 (Figure 6a). The SA-BC model severely over predicts drag at the top corner of the drag bucket at a Reynolds number of 100,000. Here the drag prediction looks similar to that at a Reynolds number of 60,000 where the model predicts no turbulent reattachment of the flow after transition.

The cause of the discrepancy in lift and drag coefficients between numerical and experimental data is explored via the distribution of the pressure coefficient along the upper and lower airfoil surface.

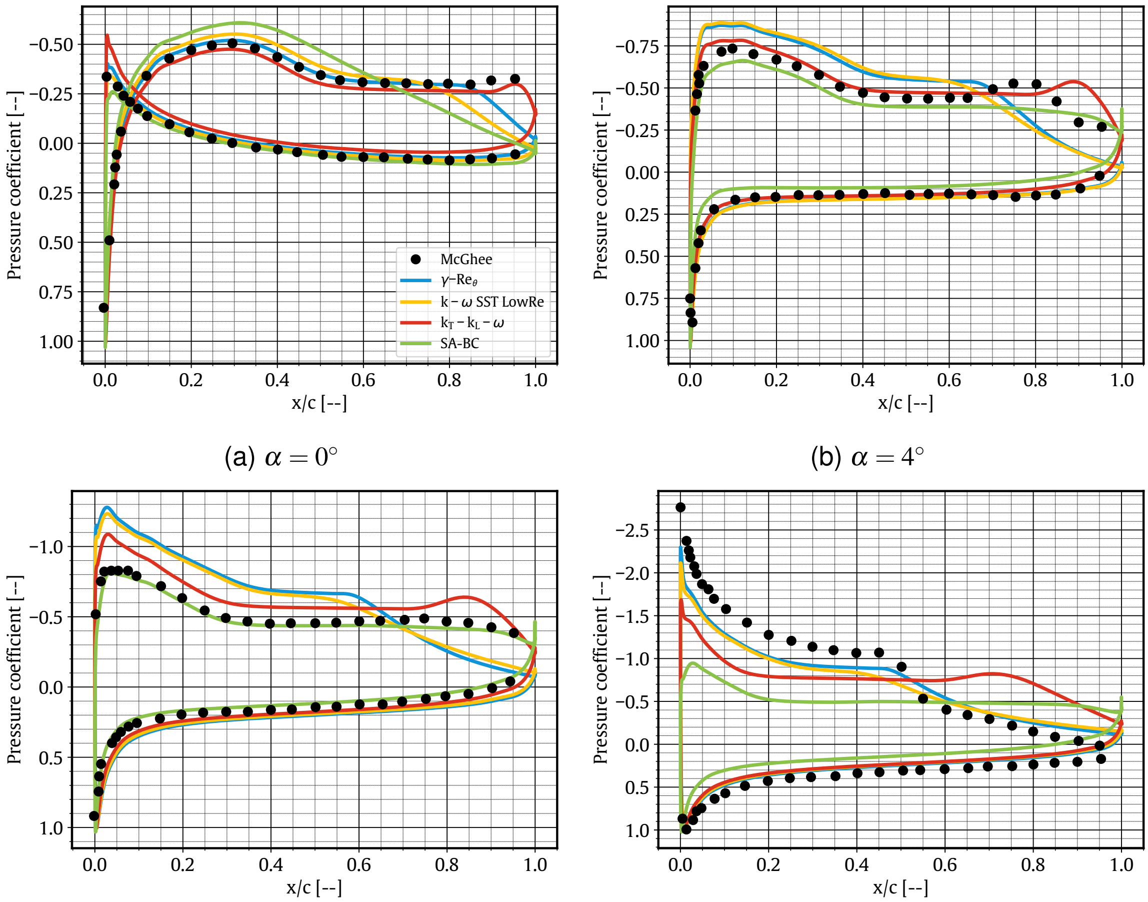


(a) Reynolds number: 100,000

(b) Reynolds number: 200,000

Figure 6 – Drag polar comparisons between the transition models and experimental data for the E387 airfoil [66, 68, 78, 80]

Figure 7 presents the coefficient of pressure along the surface of the airfoil at a Reynolds number of 60,000 and angle of attack of $0°$, $4°$, $6°$ and $8°$.



(a) $\alpha = 0°$ (b) $\alpha = 4°$

(c) $\alpha = 6°$ (d) $\alpha = 8°$

Figure 7 – Coefficient of pressure for the E387 airfoil at a Reynolds number of 60,000 and various angles of attack [66]

The experimental results by McGhee [66] show a potential transition point around 95% of the chord and no reattachment at $0°$ angle of attack and a Reynolds number of 60,000 (Figure 7a). The $k_T - k_L - \omega$ model also shows transition at 95% of the chord and no reattachment. The $\gamma - Re_\theta$ model predicts transition around 85% of the chord but predicts the pressure the best across the rest of the airfoil. The low Reynolds $k-\omega$ SST model predicts a similar pressure distribution to the $\gamma - Re_\theta$ model and shows transition around 80% of the chord. Interestingly, the intermittency function in the SA-BC model has not triggered transition at this condition as can be seen by the constant pressure coefficient gradient from around 40% of the chord to the trailing edge. This pressure distribution looks similar to results produced by a turbulence model, where there is a constant pressure coefficient gradient after the pressure peak, and not a transition model [55].

While a reasonable agreement is obtained for most of the transition models at $0°$, more substantial differences occur at $4°$ (Figure 7b). All four transition models show laminar separation around 40% of the chord, which matches the experimental results by McGhee [66]. The experiment shows transition around 80% of the chord, whereas the low Reynolds $k-\omega$ SST model and the $\gamma - Re_\theta$ model predict transition at 65% and 70% of the chord respectively. The overestimation of the pressure peak and the underestimation in the size of the laminar separation bubble from these two models is reflected in the small overestimation in lift and small underestimation in drag at $4°$ (Figure 5). The SA-BC model shows full separation at $4°$, which leads to the underprediction of lift and overprediction of drag at this condition (Figure 5). The $k_T - k_L - \omega$ model matches the experimental data the closest at $4°$, although the transition point is further aft which can explain the overestimation of drag at this condition (Figure

5b).

The experimental data by McGhee [66] presented in Figure 7b shows a decrease in the pressure coefficient from 70% to 85% of the chord. This decrease in pressure around the transition point has been seen in a Particle Image Velocimetry study conducted by Peng [84] and an unsteady RANS and DNS study conducted by Lei [85]. The behaviour is believed to have originated from a trailing edge laminar separation bubble, as described by Peng [84]. The phenomena is extremely unsteady and causes a vortex shedding behaviour from the trailing edge [84–86]. Lei [86] showed that the formation of vortices can be seen in time-averaged coefficient of pressure plots via a change in pressure from a smooth contour to a bulge or oscillatory contour. This behaviour can be seen in the $k_T - k_L - \omega$ model at all angles of attack shown in Figure 7, although the model is run in a steady RANS solver it still appears to have this time-averaged vortex behaviour.

The predictions from the SA-BC transition model match the experimental data very well at 6° angle of attack and a Reynolds number of 60,000 (Figure 7c). At this condition the flow is fully separated which is reflected in the large increase in drag and decrease in lift as shown in Figure 5. The $k_T - k_L - \omega$ model compares well to the published results at 6°, although the pressure peak is overestimated and there is a small "bulge" in the pressure distribution at the trailing edge which, as discussed earlier, indicates the presence of an unsteady vortex structure. This results in an overestimation of lift (Figure 5a). Finally, the low Reynolds $k - \omega$ SST model and the $\gamma - Re_\theta$ model both predict laminar separation to occur about 10% of the chord downstream from the experiments, and predict transition further upstream than measured. Both models do not predict full separation and show a laminar separation bubble at this condition. As a consequence, both models overestimate lift and underestimate drag at 6° angle of attack and a Reynolds number of 60,000 (Figure 5b).

At 8° angle of attack and a Reynolds number of 60,000, a laminar separation bubble is now evident in the published results by McGhee [66]. Laminar separation occurs at 20%, transition occurs at 45%, and reattachment[4] occurs at 55% of the chord (Figure 7d). All transition models predict laminar separation within 5% of the experimental results. The $\gamma - Re_\theta$ model matches the transition location and the low Reynolds $k - \omega$ SST model is within 5% of the published results. Reattachment for the two models occurs extremely close to the trailing edge which, combined with the underprediction of the magnitude of the pressure coefficient on the upper surface, leads to lower lift and higher drag than the experiment (Figure 5). The SA-BC model shows full separation and inaccurate results in lift and drag. The $k_T - k_L - \omega$ model shows transition around 80% of the chord and reattachment right on the trailing edge, which results in a underprediction in lift and overprediction in drag.

While the $\gamma - Re_\theta$ and low Reynolds $k - \omega$ SST models overall predict drag well at a Reynolds number of 100,000 (Figure 6), accuracy starts to deteriorate for the upper corner of the drag bucket. To investigate the cause of this, Figure 8 presents the pressure distribution at a Reynolds number of 100,000 and 10° angle of attack. As shown, the $\gamma - Re_\theta$ model predicts transition on the top surface of the airfoil at around 12% of the chord length whereas in the experiment [66], the transition is found to occur at 6% of the chord length. This overestimation in the size of the laminar separation bubble causes the CFD simulations to over-predict drag at high lift coefficients. On the other hand, the low Reynolds $k - \omega$ SST model predicts the size and locations of the laminar separation bubble extremely well. However, the model still over-predicts drag at this high angle of attack/lift coefficient (Figure 6a). This over-prediction of drag can be attributed to the over estimation of C_p at the leading edge.

The locations of laminar separation, transition, and turbulent reattachment of the E387 airfoil predicted by the four transition models and by Xfoil are compared to experimental data in Figures 9 and 10 [77, 87].

As shown in Figures 9a and 9b, the SA-BC transition model generally predicts the laminar separation and transition locations well for the E387 airfoil at a Reynolds number of 100,000. However, the model predicts turbulent reattachment locations further downstream than experimental results. The

[4]Accurate predictions of reattachment from a coefficient of pressure plot is difficult, the reattachment predictions for the CFD models are completed by analysing the skin friction coefficient.

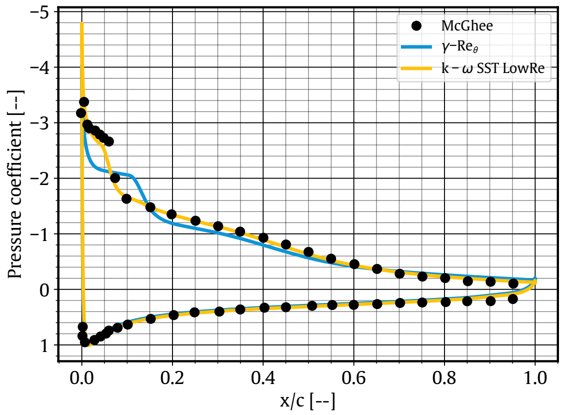
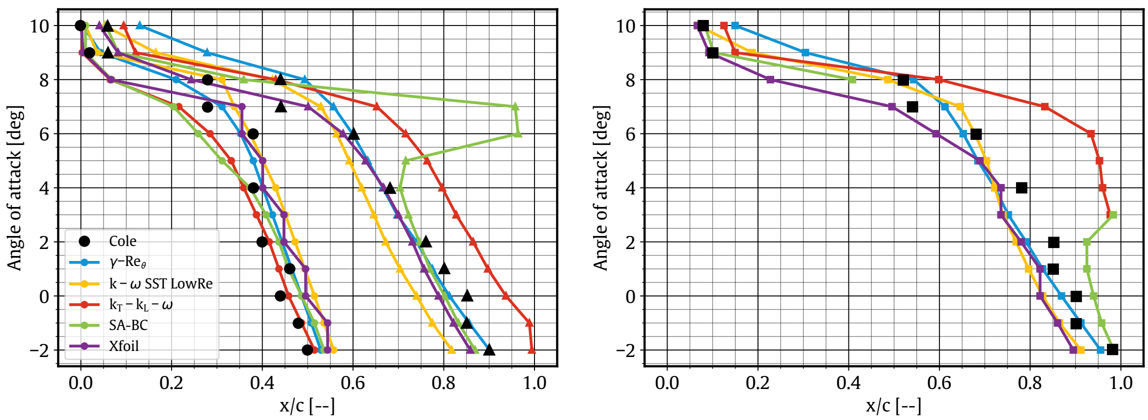


Figure 8 – Coefficient of pressure of the E387 airfoil at a Reynolds number of 100,000 and angle of attack of 10° with experimental data [66]



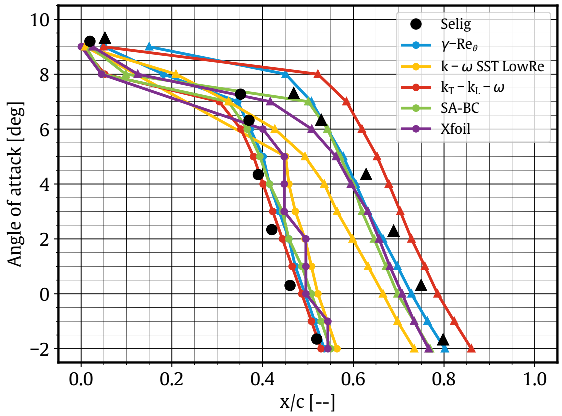
(a) Laminar separation and transition

(b) Turbulent reattachment

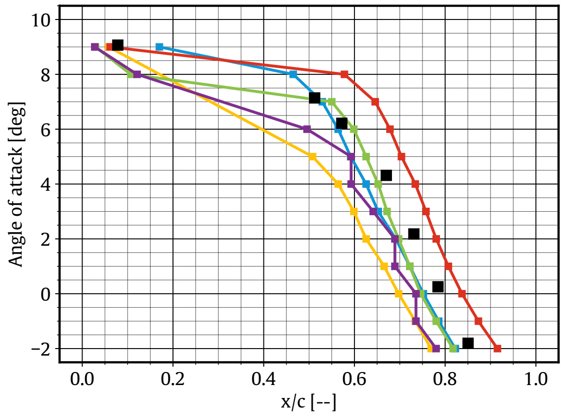
Figure 9 – Laminar separation (●), transition (▲) and turbulent reattachment (■) locations of the E387 airfoil alongside experimental data at a Reynolds number of 100,000 [77]

$k_L - k_T - \omega$ transition model also predicts the transition and turbulent reattachment locations further aft and fails to capture turbulent reattachment from -2° to 2° angles of attack as indicated by the missing data points on Figure 9b. The SA-BC model also does not capture turbulent reattachment from 4° to 7° angles of attack (which explains the gap in the SA-BC data in Figure 9b). The fully separated flow for the SA-BC model is reflected in Figure 6a, where the drag increases drastically at these angles of attack. Xfoil presents excellent agreement with experimental data and captures the rapid forward shift in all locations at higher angles of attack. It does, however, predict turbulent reattachment slightly upstream of the experimental locations. The $\gamma - Re_{\theta}$ transition model predicts the laminar separation, transition and turbulent reattachment locations closest to the experimental values. Predictions for separation locations generally fall within 10% chord, while transition and turbulent reattachment locations show an average difference of less than 6% chord. While the $\gamma - Re_{\theta}$ model does capture the rapid forward shift in locations with increasing angle, the predicted shift is slightly slower than the experimental study which explains why drag in the upper corner of the drag bucket is not predicted as well (Figure 6a). The low Reynolds number $k - \omega$ SST model predicts laminar separation, transition and turbulent reattachment well at higher angles of attack. At low angles, separation is predicted further aft and transition and turbulent reattachment is predicted early. As this forms a smaller laminar separation bubble, drag is under-predicted near the bottom corner of the drag bucket (Figure 6a).

As expected, all models generally perform better at a Reynolds number of 200,000 (Figures 10a and 10b). The most notable improvement occurs for the SA-BC and $k_T - k_L - \omega$ models. Whereas both



(a) Laminar separation and transition



(b) Turbulent reattachment

Figure 10 – Laminar separation (●), transition (▲) and turbulent reattachment (■) locations of the E387 airfoil alongside experimental data at a Reynolds number of 200,000 [87]

models failed to predict reattachment at the lower Reynolds number and predicted transition poorly, they now align much better with the experimental data. The SA-BC transition model predicts the length of the laminar separation bubble (laminar separation to turbulent reattachment) very closely to the experimental data at a Reynolds number of 200,000. This correlates to a very good prediction of drag as seen in Figure 6b. Except for the low Reynolds number $k-\omega$ SST model, all transition models predict separation locations better than Xfoil which tends to predict this too far downstream. Most models, except for the $k_T - k_L - \omega$, predict transition and reattachment early. As Xfoil predicts separation to occur further downstream and transition and reattachment upstream of the measured value, a smaller laminar separation bubble is produced. This can explain why Xfoil tends to under-predict drag [47, 88]. The average errors in predicting the laminar separation, transition, and turbulent reattachment produced by the $\gamma - Re_\theta$ model are 4.6%, 3.7% and 5.4% of the chord respectively. As before, the low Reynolds $k-\omega$ SST model predicts transition and turbulent reattachment upstream and laminar separation downstream of the measured values between -2° and 5° angle of attack. At angles of attack between 6° and 8°, the low Reynolds $k-\omega$ SST model shows no separation, although the flow does transition (Figure 10). The flow undergoes natural transition further forward than the experimental separation-induced transition point at these angles of attack.

The $\gamma - Re_\theta$ model tends to predict the transition and reattachment location upstream of the experimental values below 7° angles of attack as presented in Figure 9 and 10. The $\gamma - Re_\theta$ model uses an effective intermittency feature to trigger faster flow reattachment after transition in separation-induced transition [57, 60]. This will allow the flow to reattach very shortly after the transition point [34]. As stated by Langtry [57], the transport equations do not attempt to model the physics of the transition process (unlike turbulence models) but instead form a framework for implementation of correlation based models into general-purpose CFD methods. As the model relies on experimental correlations to determine the length of the transition region, it will not always predict this transition point accurately [34]. On the other hand, the $k_T - k_L - \omega$ model consistently predicts both transition and turbulent reattachment further aft than experimental data (Figure 9 and 10). The $k_T - k_L - \omega$ model was developed as a phenomenological, or physics-based model that tries to model the transition related phenomena [34]. The physics of transition is non-linear and highly unsteady, therefore a steady RANS $k_T - k_L - \omega$ simulation can not accurately predict this transition component of the flow.

The low Reynolds $k-\omega$ SST and the $\gamma - Re_\theta$ models predict lift performance similarly, whereas the prediction of drag for the two models diverges with an increase in lift coefficient at a Reynolds number of 100,000 (Figure 6a). Interestingly, the laminar separation and reattachment points for these two models at this condition are quite close, which indicates a laminar separation bubble of very similar length (Figure 9). The flow fields of these two models as well as the other two transition models at a Reynolds number of 100,000 and at 7° angle of attack are further analysed to get an indication for the cause of the differences in predicted drag. Figure 11 shows the turbulent kinetic energy of the flow as well as velocity streamlines for the $\gamma - Re_\theta$, low Reynolds $k-\omega$ SST, and the $k_T - k_L - \omega$ models. As the transport equations for the SA-BC model are different, the kinematic eddy turbulent viscosity is shown instead. The $\gamma - Re_\theta$ and the low Reynolds $k-\omega$ SST models show a laminar separation bubble length of 30% of the chord with transition occurring 25% and 20% aft of the separation point for the $\gamma - Re_\theta$ and low Reynolds $k-\omega$ SST models respectively (Figure 9a). The physics of transition by the $\gamma - Re_\theta$ model is contained purely in the experimental correlations within the model [60]. In a similar manner, the low Reynolds $k-\omega$ SST model has been designed to guarantee that the point at which the turbulent kinetic energy is first amplified matches the minimum critical Reynolds number for an incompressible flat-plate boundary layer [56]. As explained in Section 2, the two models use different mechanisms for transition which lead to different laminar separation bubble heights and therefore different drag coefficients (Figure 11).

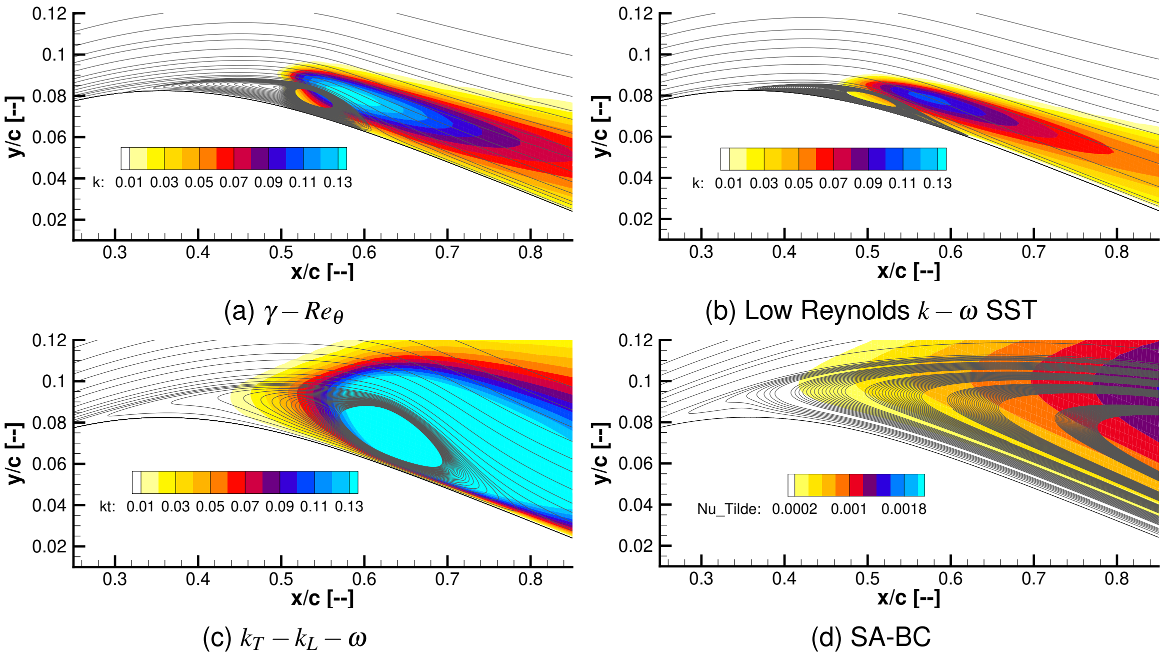


Figure 11 – Turbulent kinetic energy with velocity streamlines of the E387 airfoil at 7° angle of attack and a Reynolds number of 100,000

The $k_T - k_L - \omega$ model produces a longer laminar separation bubble at a Reynolds number of 100,000 and at 7° angle of attack (Figure 9). This can be seen in Figure 11c where the laminar flow separates further upstream and reattaches further downstream than the $\gamma - Re_\theta$ and the low Reynolds $k-\omega$ SST models. Flow predicted with the SA-BC model, on the other hand, separates but does not reattach (Figure 11d).

The $k_T - k_L - \omega$ model has the largest distance from separation to the onset of transition (where the turbulent kinetic energy begins to grow). This leads to a longer and potentially deeper laminar separation bubble. The region between separation and the onset of transition is the region that the $k_T - k_L - \omega$ model attempts to capture in a physical manner. Although, the model has no specific terms for separation-induced transition, it implicitly assumes that the transition process is comparable between attached and separated boundary layers [58, 61]. Separation-induced transition, at a low turbulence intensity, is similar to natural transition although there are additional instabilities including the Kelvin-Helmholtz instability which has been found to be dominant in the transition process [59, 89]. The omission of terms for separation-induced transition in the $k_T - k_L - \omega$ model could be one of the contributing factors to the overestimation of the laminar separation bubble length and therefore the drag coefficient by the $k_T - k_L - \omega$ model. As seen in Figure 11d, the SA-BC model has the largest separation angle (the angle between the dividing streamline, which is the line between the forward moving flow and the recirculation zone, and the airfoil surface) which causes the flow to separate without reattachment. This is an issue for the SA-BC model at a Reynolds number of

100,000 between 4° and 7° angle of attack (Figure 9).

The upper surface skin friction at a Reynolds number of 100,000 and 7° angle of attack is used to further explore differences between the various transition models (Figure 12). As stated previously, the skin friction coefficient can be used to identify key bubble-related positions, and, in conjunction with Figure 11, helps to understand the prediction of the four transition models at this condition. Figure 12 shows that the $\gamma - Re_\theta$ model reattaches shortly after transition, whereas flow predicted by the low Reynolds $k - \omega$ SST model takes longer to reattach after transition (Figure 12). The $\gamma - Re_\theta$ models' use of an effective intermittency for separation-induced transition can be seen in Figures 11a and 12, where the production of the turbulent kinetic energy is accelerated thus forcing reattachment shortly after the onset of transition. The low Reynolds $k - \omega$ SST model shows a weaker production of turbulent kinetic energy (Figure 11b), which results in a slower reattachment of the flow as shown in Figure 12. The weaker production of turbulent kinetic energy by the low Reynolds $k - \omega$ SST model may be sufficient for some conditions and not for others. As damping functions are calibrated and optimised for certain conditions, in this case an incompressible flat plate, some issues may arise when trying to predict transitional behaviour in the flow [56].

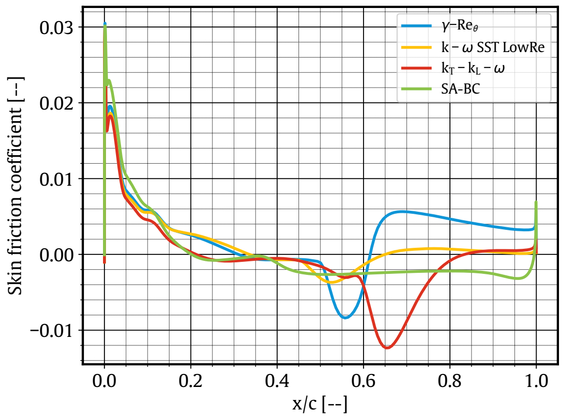
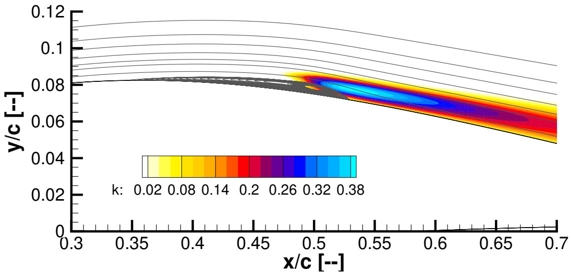


Figure 12 – Upper surface skin friction coefficient of the E387 airfoil at a Reynolds number of 100,000 and 7° angle of attack

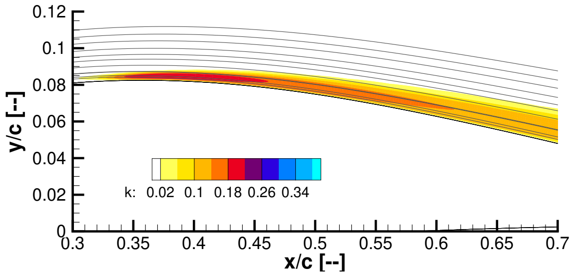
The $k_T - k_L - \omega$ model again shows early separation and a larger length between separation and transition onset, when compared to the $\gamma - Re_\theta$ and low Reynolds $k - \omega$ SST models (Figure 12). As stated previously, this could be due to the model not having terms for separation-induced transition [34, 58]. As can be seen in Figure 11c, the early separation and delayed onset of transition causes the dividing streamline to move further away from the surface of the airfoil before the onset of transition. This makes it more difficult for the flow to reattach once it has transitioned and, consequently, causes a longer and higher laminar separation bubble. The SA-BC models prediction of length from separation to the onset of transition is about 3% shorter than the $k_T - k_L - \omega$ model. As can be seen in Figure 11d and 12, the SA-BC model has a slow production of turbulence and therefore a large transition length. The SA-BC model does not have a transition length correlation and is calibrated against a zero pressure gradient flat plate test case. This could be an issue with the SA-BC model when using it at certain flow conditions. It appears that these issues are minimised at Reynolds numbers at or above 200,000.

The flow fields of the four transition models at a Reynolds number of 200,000 and at 7° angle of attack are further analysed to see the change in the bubble predictions at a higher Reynolds number (Figure 13). The skin friction and pressure coefficients are also presented at this condition in Figure 14.

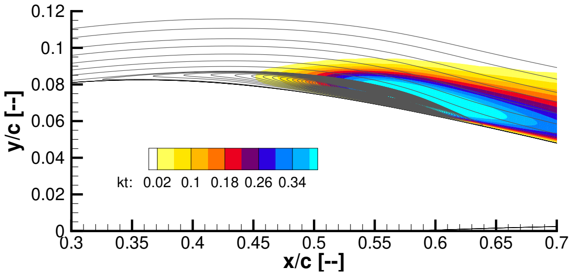
The $\gamma - Re_\theta$ and the SA-BC models predict separation, transition and reattachment at very similar locations at a Reynolds number of 200,000 and at 7° angle of attack (Figure 10). Figure 14b shows the coefficient of pressure peak for the SA-BC model is larger than the $\gamma - Re_\theta$ model, although the pressure around the rest of the airfoil is much the same. The laminar separation bubble for



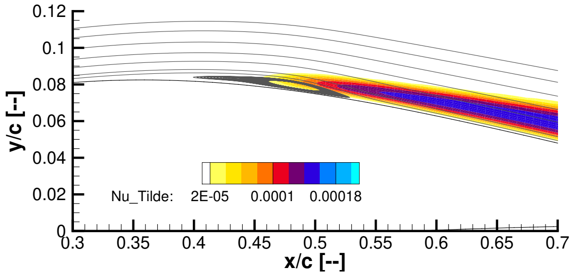
(a) $\gamma - Re_\theta$



(b) Low Reynolds $k-\omega$ SST



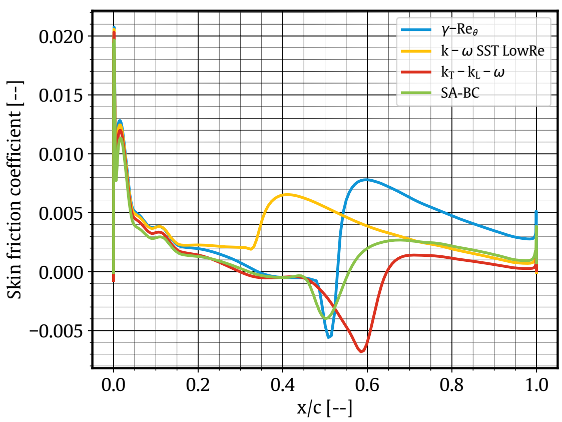
(c) $k_T - k_L - \omega$



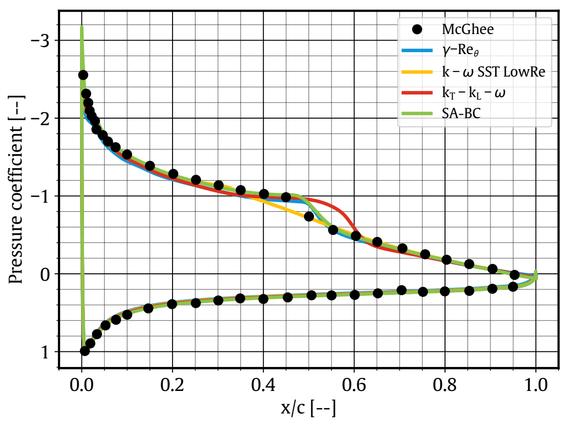
(d) SA-BC

Figure 13 – Turbulent kinetic energy with velocity streamlines of the E387 airfoil at 7° angle of attack and a Reynolds number of 200,000

these two models also look very similar although there is some differences in the reattachment zone. Figure 13 shows the $\gamma - Re_\theta$ model has a rapid reattachment of the flow whereas the reattachment with the SA-BC model is smoother. This rapid reattachment appears to increase the skin friction coefficient aft of reattachment as seen in Figure 14a. The larger skin friction coefficient will lead to a larger coefficient of drag as seen in Figure 6b, where the $\gamma - Re_\theta$ model overpredicts drag at this condition. The $k_T - k_L - \omega$ model shows similar results at 7° angle of attack and a Reynolds number of 100,000, where the length from separation to transition onset is larger than the other models showing separation. Figure 14a shows the $k_T - k_L - \omega$ model has the largest transition to reattachment length which leads to a smoother reattachment of the flow as seen in Figure 13c. These factors lead to a longer and higher laminar separation bubble for the $k_T - k_L - \omega$ model which will create an overestimation in the pressure drag on the airfoil and in turn a larger coefficient of drag than published results. At a Reynolds number of 200,000 and at 7° angle of attack the low Reynolds $k-\omega$ SST model does not separate and undergoes natural transition as seen in Figure 13b and Figure 14a where the friction coefficient stays positive.



(a) Upper surface skin friction coefficient



(b) Coefficient of Pressure

Figure 14 – Skin friction coefficient and coefficient of pressure of the E387 airfoil at a Reynolds number of 200,000 and 7° angle of attack

4.2 Wortmann FX63-137

The Wortmann FX63-137 is a 13.7% thick airfoil with a maximum camber of 6% and is designed to produce high lift at low Reynolds numbers. This section will present the aerodynamic performance of the FX63-137 airfoil at Reynolds numbers of 100,000, 150,000 and 200,000, as well as the laminar separation bubble locations at a Reynolds number of 200,000. The airfoil shape is shown in Figure 15 while performance predictions for a Reynolds number of 100,000 are given in Figure 16.

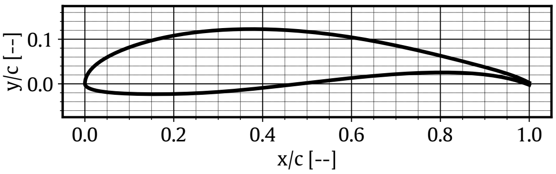
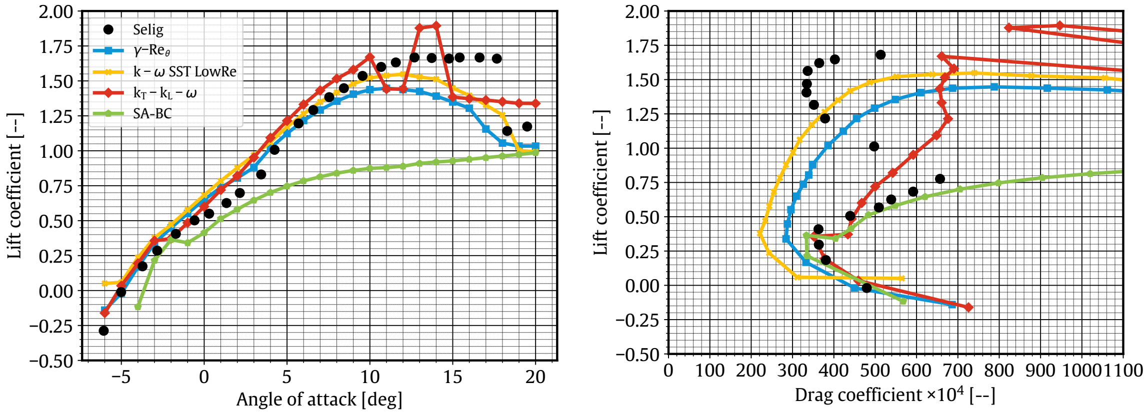


Figure 15 – FX63-137 airfoil



(a) Angle of attack and coefficient of lift

(b) Lift coefficient versus drag coefficient

Figure 16 – Comparisons between the transition models and experimental data of the FX63-137 airfoil at a Reynolds number of 100,000 [68]

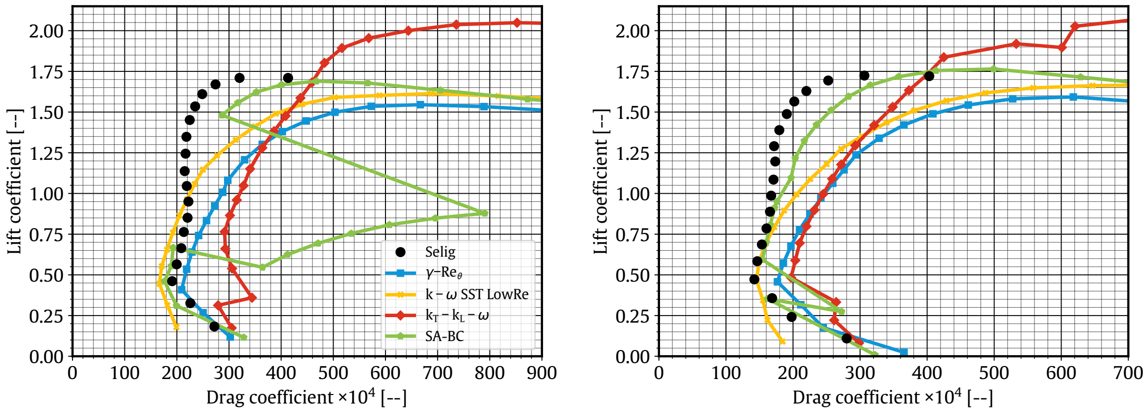
The SA-BC model produces lower lift coefficients across the entire range of angles of attack, similar to the previous airfoils at a Reynolds number of 60,000 (Figure 16a). Interestingly, all of the transition models (apart from SA-BC) over-predict lift between the angles of attack of -0.5° and 4.5° where a slight drop of lift curve slope is observed in the experimental data. The maximum lift coefficient is under-predicted expect the $k_T - k_L - \omega$ model, along with the stall angle for all models. While an abrupt stall behaviour is found in the experiment, except for the $k_T - k_L - \omega$ model, none of the transition models manage to capture this. The $k_T - k_L - \omega$ model appears to have a false prediction in stall, where an initial stall is predicted followed by another stall 4° later. This is because the model is predicting full separation at 11°-12°, then the reattachment of the flow again at 13°. The flow around this area is unsteady and therefore the steady RANS model struggles to predict the aerodynamic performance accurately.

The drag polar of the FX63-137 airfoil at a Reynolds number of 100,000 shows the large reduction in drag from moderate lift coefficients prior to stall (Figure 16b). As the flow is potentially fully separated at moderate lift coefficients (as stated previously for the E387 at a Reynolds number of 60,000), the SA-BC and $k_T - k_L - \omega$ models match the experimental drag values the closest at this condition. The other models show no sign of capturing this rapid increase in drag at moderate lift coefficients. As shown, none of the presented transition models are able to capture the drag at high lift coefficients accurately.

The performance of the transition models are investigated at Reynolds numbers of 150,000 and 200,000 (Figure 17). Drag is poorly predicted for all the models at a Reynolds number of 150,000.

With the exception of the low Reynolds $k-\omega$ SST at low lift coefficients, all models consistently over-predict drag at moderate to high lift coefficients. The $\gamma-Re_\theta$, low Reynolds $k-\omega$ SST and SA-BC transition models are able to capture the drag values quite accurately at the low lift coefficients. On the other hand, a slight improvement in drag prediction is observed for the Reynolds number of 200,000 (Figure 17b).

As can be seen in the drastic increase in drag at a Reynolds number of 150,000, the SA-BC model predicts full laminar separation at moderate lift coefficients, before reattachment occurs at high lift coefficients. Interestingly, a drastic improvement in drag prediction is found for the SA-BC model at a Reynolds number of 200,000. Here, the drag polar matches experimental values accurately. It is interesting to see the progression of the SA-BC model from Reynolds numbers of 100,000 to 150,000 and finally to 200,000 where the flow is fully separated, to partially separated, to fully reattached across all angles of attack within the drag bucket (Figures 16b, 17a and 17b). The increase in Reynolds number leads to a lot better aerodynamic predictions from the SA-BC model.



(a) Reynolds number: 150,000

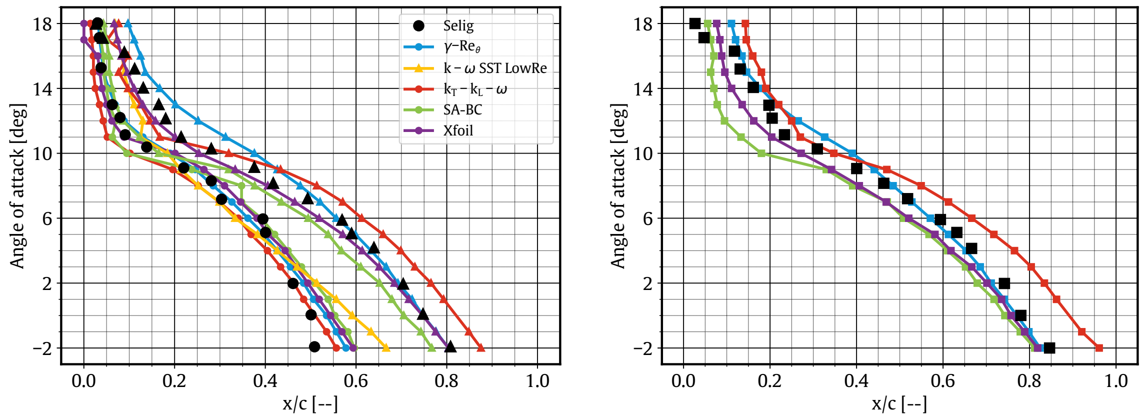
(b) Reynolds number: 200,000

Figure 17 – Drag polar comparisons between the transition models and experimental data for the FX63-137 airfoil [39]

The separation, transition, and reattachment locations by the four transition models and Xfoil for the FX63-137 airfoil at a Reynolds number of 200,000 are given in Figure 18. The predicted transition locations generally match experimental data well (Figure 18). Similar to the E387 airfoil a Reynolds number of 200,000, the SA-BC model predicts an earlier transition and reattachment, whereas the $k_T-k_L-\omega$ predicts both locations further downstream than experimental measurements at angles below 10°. Once again, this will form a longer laminar separation bubble and cause the over-prediction of drag shown in Figure 17b. At higher angles of attack, the $k_T-k_L-\omega$ turbulence model predicts the separation and transition locations forward and the reattachment point aft of the experimental data. Xfoil, once more, performs well for all locations but tends to predict transition and reattachment locations slightly upstream of the recorded locations. The $\gamma-Re_\theta$ model, once again, predicts the transition and reattachment locations best although its performance slightly degrades at higher angles of attack. The low Reynolds $k-\omega$ SST model shows no flow separation for the FX63-137 airfoil at a Reynolds number of 200,000 and again predicts an early transition, especially at low angles of attack.

The flow fields of the four transition models at a Reynolds number of 200,000 and at 5° angle of attack are further analysed to see the differences in the bubble predictions for the FX63-137 airfoil (Figure 19). This condition was chosen as the laminar separation bubble lengths are similar but the aerodynamic performance is quite different. The skin friction and pressure coefficients are also presented at this condition in Figure 20.

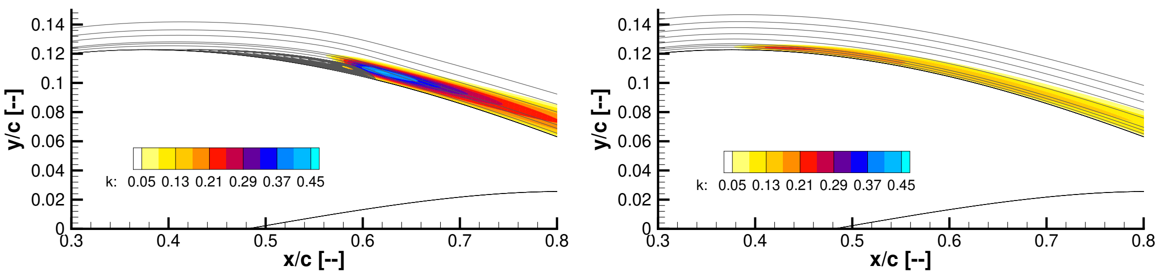
The SA-BC and the $\gamma-Re_\theta$ models both have flow separation around 40% of the chord but the SA-BC model transitions and reattaches earlier than the $\gamma-Re_\theta$ model (Figure 20a). Figure 19 shows that this results in a similar laminar separation bubble height for the two models up until the



(a) Laminar separation and transition

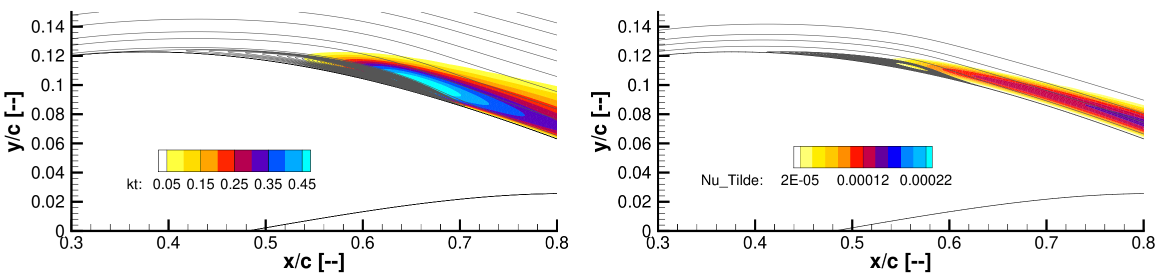
(b) Turbulent reattachment

Figure 18 – Laminar separation (●), transition (▲) and turbulent reattachment (■) locations of FX63-137 airfoil alongside experimental data at a Reynolds number of 200,000 [87]



(a) $\gamma - Re_\theta$

(b) Low Reynolds $k-\omega$ SST

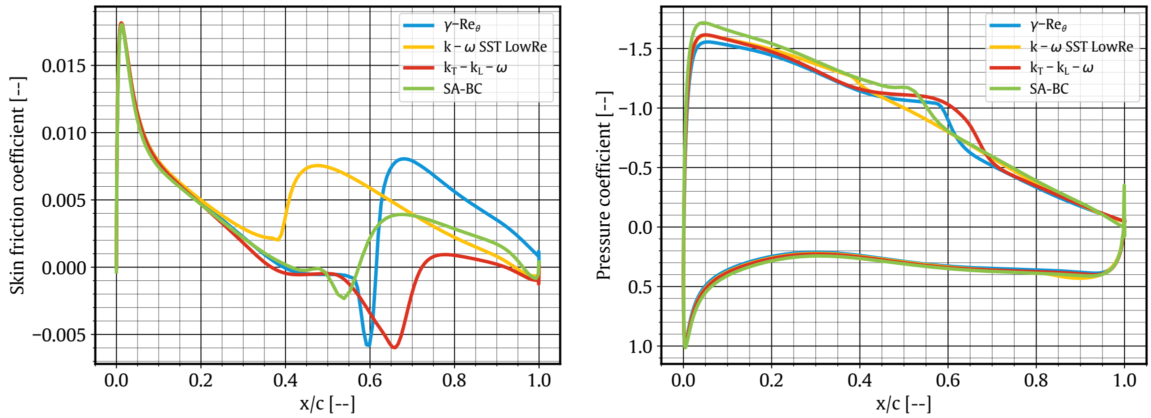


(c) $k_T - k_L - \omega$

(d) SA-BC

Figure 19 – Turbulent kinetic energy with velocity streamlines of the FX63-137 airfoil at 5° angle of attack and a Reynolds number of 200,000

transition point of the SA-BC model. The longer and higher laminar separation bubble from the $\gamma - Re_\theta$ model aft of 55% of the chord, will result in a larger drag coefficient. The $\gamma - Re_\theta$ model also shows a rapid reattachment after transition which results in the larger skin friction peak after reattachment, whereas the SA-BC model shows a moderate skin friction peak after reattachment which would lead to a smaller drag coefficient. The SA-BC model is closest to published drag results while overestimating lift by the largest amount for the four transition models at this condition (Figure 17b). The overestimation in lift can be attributed to the larger pressure coefficient distribution across the upper surface of the airfoil (Figure 20b). The $k_T - k_L - \omega$ model shows the longest length from transition onset to reattachment which results in a longer laminar separation bubble and the smallest peak skin friction after reattachment (Figure 19c & 20a). The longer bubble can be attributed to the slower production of turbulent kinetic energy in comparison to the $\gamma - Re_\theta$ model. The larger laminar separation bubble for the $k_T - k_L - \omega$ model, which results in the larger pressure bulge around transition in Figure 20b, will result in the largest pressure drag increase for the four models due to the laminar separation bubble. Interestingly, both the $k_T - k_L - \omega$ model and the $\gamma - Re_\theta$ model have almost identical drag coefficients with quite different looking laminar separation bubbles and in turn skin friction and pressure distributions. The $\gamma - Re_\theta$ model would have a larger portion of drag due to skin friction compared to the $k_T - k_L - \omega$ model, whereas the $k_T - k_L - \omega$ model would have a larger portion of drag due to pressure than the $\gamma - Re_\theta$ model. The Low Reynolds $k - \omega$ SST model shows no separation and a natural transition around 40% of the chord at this condition (Figure 19b). The premature transition to a turbulent flow at this condition causes an increase in skin friction, and in turn an increase in the drag coefficient.



(a) Upper surface skin friction coefficient

(b) Coefficient of Pressure

Figure 20 – Skin friction coefficient and coefficient of pressure of the FX63-137 airfoil at a Reynolds number of 200,000 and 5° angle of attack

4.3 NACA0018

The NACA0018 airfoil is a symmetrical airfoil with a maximum thickness of 18% and is typically used in wind energy applications and vertical axis wind turbines [90]. This section will present the aerodynamic performance of the NACA0018 airfoil at a Reynolds number of 150,000 as well as the laminar separation bubble locations at a Reynolds number of 100,000 and 150,000. The aerodynamic performance and the laminar separation bubble simulations were run at two different turbulence intensities to match the experimental data available. The aerodynamic performance simulations were run at a turbulence intensity of 0.02% to match Timmer [71] and the laminar separation bubble simulations were run at a turbulence intensity of 0.19% to match the experimental data by Boutilier [72]. The airfoil shape is shown in Figure 21 and the aerodynamic performance for a Reynolds number of 150,000 is depicted in Figure 22. The published lift coefficients by Boutilier [72] were also included in Figure 22a, although no drag data was available and therefore is omitted from Figure 22b.

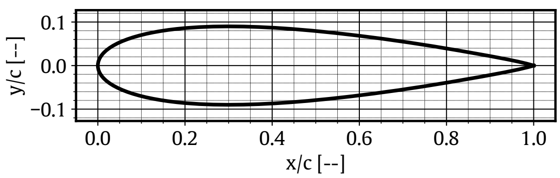
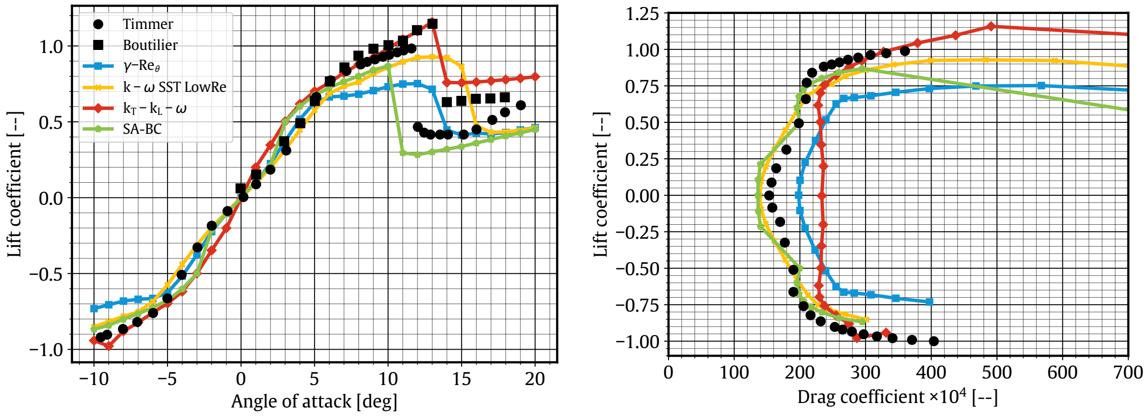


Figure 21 – NACA0018 airfoil

The two published lift values are similar, although they do differ around stall which can be attributed to the different turbulence intensities and setups of the two tests. The lift values predicted by the SA-BC model match the experimental data well up until stall, where the stall angle and maximum lift coefficient are under-predicted (Figure 22a). Similarly, the $\gamma - Re_\theta$ model also under-predicts the maximum lift coefficient while the $k_T - k_L - \omega$ model over-predicts the maximum lift coefficient and stall angle. However, the $k_T - k_L - \omega$ model does match the experimental data by Boutilier [72] around stall. This shows the model may experience some sort of artificial turbulence at these conditions. The low Reynolds number $k - \omega$ SST model predicts the maximum lift coefficient the closest to experimental



(a) Angle of attack and coefficient of lift

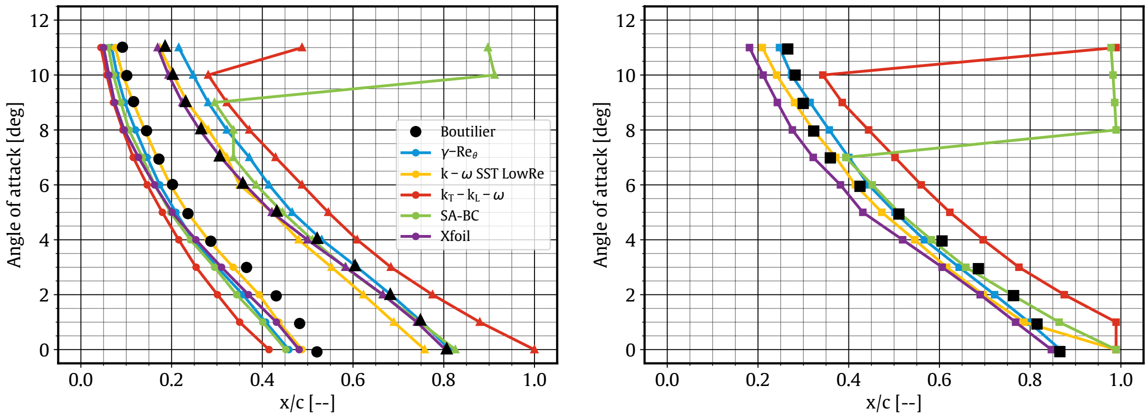
(b) Lift coefficient versus drag coefficient

Figure 22 – Comparisons between the transition models and experimental data of the NACA0018 airfoil at a Reynolds number of 150,000 [71]

data by Timmer [71], however, its stall angle is over-predicted. The experimental values indicate that an abrupt stall occurs, which all transition models can predict.

The drag polar of the NACA0018 for a Reynolds number of 150,000 is depicted in Figure 22b. The $k_T - k_L - \omega$ and the $\gamma - Re_\theta$ models over-predict drag. While the low Reynolds $k - \omega$ SST and SA-BC models slightly under-predict drag values at moderate lift coefficients, the trends of the drag polar are captured the best by these two transition models.

The laminar separation bubble predictive capability of the transition models aswell as Xfoil is shown in Figures 23 and 24 for the NACA0018 airfoil at a Reynolds number of 100,000 and 150,000 respectively.



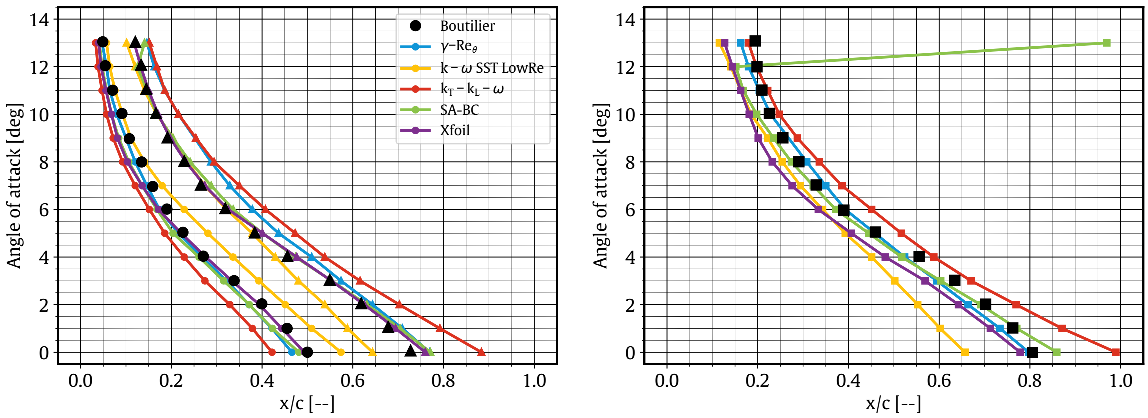
(a) Laminar separation and transition

(b) Turbulent reattachment

Figure 23 – Laminar separation (●), transition (▲) and turbulent reattachment (■) locations of NACA0018 airfoil alongside experimental data at a Reynolds number of 100,000 [72]

At a Reynolds number of 100,000, the SA-BC transition model predicts the laminar separation locations better than the $k_T - k_L - \omega$ transition model for the investigated angles (Figure 23). As expected, the $k_T - k_L - \omega$ model predicts transition and reattachment further downstream than experimental data. The results from the SA-BC model are very close to the $\gamma - Re_\theta$ model. However, the SA-BC model fails to predict the transition and reattachment locations accurately at higher angles of attack and shows signs of full separation. This is similar behaviour to the E387 case. Although, the SA-BC model appears to predict reattachment better than the E387 case at smaller angles of attack which could be due to the larger turbulence intensity for the NACA0018 (Figure 23). Out of all the tested models, the low Reynolds number $k-\omega$ SST model predicts the separation location the best. It also predicts the transition location better at higher angles of attack, whereas the $\gamma-Re_\theta$ model predicts transition locations better at lower angles of attack. Both the $\gamma-Re_\theta$ and low Reynolds $k-\omega$ SST models predict the reattachment locations the closest to the published data. While Xfoil predicts the location of the transition the closest to the experimental results, it consistently predicts reattachment locations further upstream at all angles of attack.

Similar trends can also be observed for a Reynolds number of 150,000, although with a slight improvement in the accuracy across all models bar the low Reynolds $k-\omega$ SST model (Figures 24a and 24b). While laminar separation and reattachment are predicted well by the $\gamma-Re_\theta$ model, transition locations are predicted downstream of the published data especially at high angles of attack. Once again, transition and turbulent reattachment locations predicted by $k_T-k_L-\omega$ model are further aft than measured experimentally. Interestingly, the SA-BC model predicts the laminar separation and transition locations the best among all transition models and predicts reattachment within 5% of the published results (except for the last angle of attack). On the other hand, the laminar separation and transition locations are predicted well by Xfoil but the turbulent reattachment prediction is poor in comparison to the transition models. The low Reynolds $k-\omega$ SST model predicts both transition and turbulent reattachment early while the laminar separation locations are predicted further downstream at low angles of attack. These predictions will result in the formation of a smaller laminar separation bubble which explains the under-prediction in the drag calculations in Figure 22b.



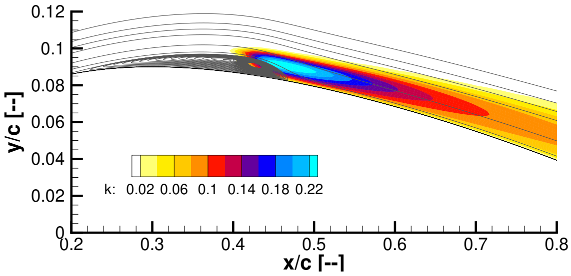
(a) Laminar separation and transition

(b) Turbulent reattachment

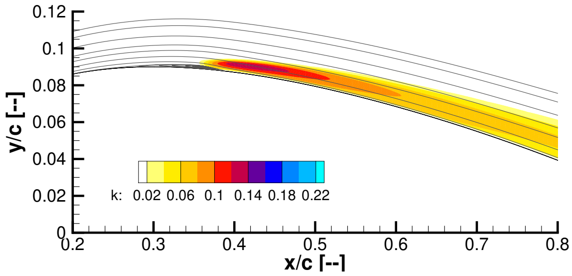
Figure 24 – Laminar separation (●), transition (▲) and turbulent reattachment (■) locations of NACA0018 airfoil alongside experimental data at a Reynolds number of 150,000 [72]

The NACA0018 airfoil flow fields of the four transition models at a Reynolds number of 150,000 and at 5° angle of attack are further analysed to see the differences in the laminar separation bubble predictions (Figure 25). This angle of attack was chosen as laminar separation bubble lengths are similar but different drag coefficients are seen at this condition. The skin friction and pressure coefficients are also presented at this condition in Figure 26.

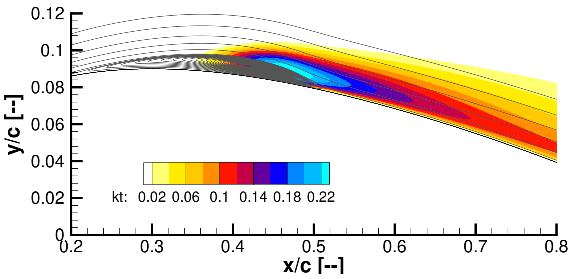
The SA-BC and the $\gamma-Re_\theta$ models show laminar separation bubbles of similar length to published results at 5° angle of attack and a Reynolds number of 150,000, whereas the Low Reynolds $k-\omega$ SST model underestimates and the $k_T-k_L-\omega$ model overestimates the length of the bubble (Figure 24). Figure 25 shows the $\gamma-Re_\theta$ model has a rapid reattachment again, whereas the SA-BC model has a smooth flow reattachment. This is also evident where the skin friction peak after reattachment is almost double for the $\gamma-Re_\theta$ model (Figure 26a). The laminar separation bubble height is also higher for the $\gamma-Re_\theta$ model, which would result in a larger drag coefficient. The $\gamma-Re_\theta$ model overestimates drag at this condition which could be from the larger peak in skin friction after reattachment and also the height of the laminar separation bubble. The SA-BC model predicts drag the closest to published results at this condition, but over-predicts lift. This could mean the laminar separation bubble is predicted accurately but the peak pressure coefficient is too large and therefore the lift coefficient is



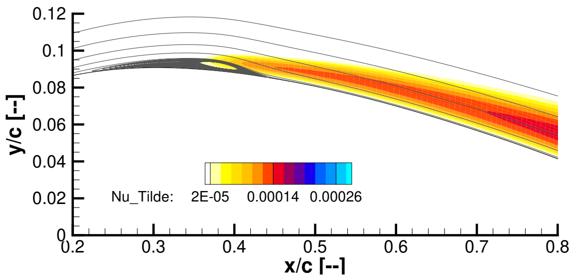
(a) $\gamma - Re_\theta$



(b) Low Reynolds $k-\omega$ SST



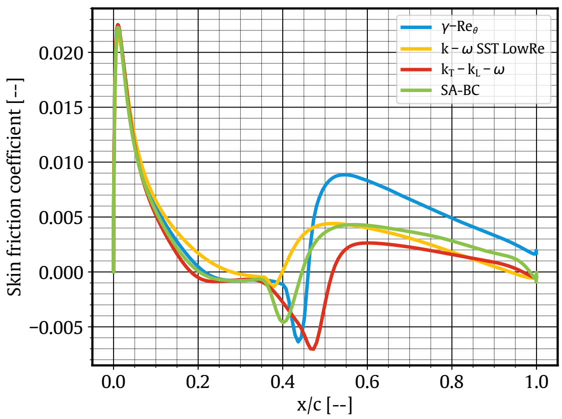
(c) $k_T - k_L - \omega$



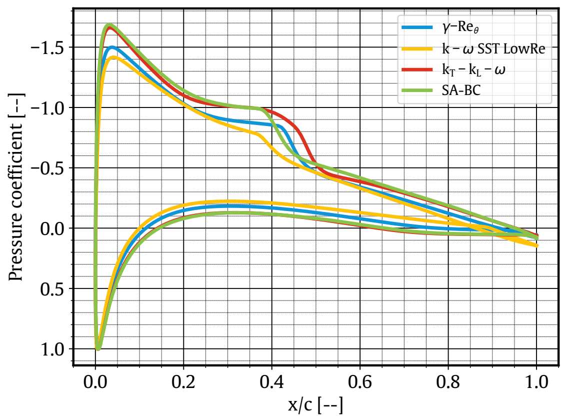
(d) SA-BC

Figure 25 – Turbulent kinetic energy with velocity streamlines of the NACA0018 airfoil at 5° angle of attack and a Reynolds number of 150,000

too large. The peak pressure coefficient of the $k_T - k_L - \omega$ model is similar to the SA-BC model and it also over-predicts lift at this condition, whereas the $\gamma - Re_\theta$ model shows a smaller peak pressure coefficient and predicts lift the most accurately (Figure 26b). The $k_T - k_L - \omega$ model shows the largest bubble height but does show a smoother flow reattachment than the $\gamma - Re_\theta$ model, this could be why the model over-predicts drag but not as much as the $\gamma - Re_\theta$ model at this condition. The Low Reynolds $k-\omega$ SST model shows the smallest laminar separation bubble as well as the smallest pressure distribution across the airfoil, this leads to an underestimation in lift and drag at this condition (Figure 25b & 26b).



(a) Upper surface skin friction coefficient



(b) Coefficient of Pressure

Figure 26 – Skin friction coefficient and coefficient of pressure of the NACA0018 airfoil at a Reynolds number of 150,000 and 5° angle of attack

It has been found throughout this study and through previous studies [48, 50, 83, 91], that at a combination of low turbulence intensities ($\leq$0.2%), low Reynolds numbers ($\leq$300,000) and at low to moderate pressure gradients, the $k_T - k_L - \omega$ transition model in a steady RANS solver is unable to capture the transition phenomena and in turn the aerodynamic performance accurately.

5. Conclusion

While the predictive capability of various transition models at Reynolds numbers around 200,000 has been investigated widely, to date no study has explored the performance of these transition models at lower Reynolds numbers of interest to small UAVs. This paper is the first to do so using open-source CFD codes with a steady RANS solver for a total of three airfoils. Aerodynamic coefficients such as lift and drag and flow feature locations (laminar separation, transition, and turbulent reattachment locations) are compared to experimental results. A total of four transition models: $\gamma - Re_\theta$, low Reynolds $k-\omega$ SST, $k_T - k_L - \omega$ and Spalart-Allmaras with transition are compared to experiments at Reynolds numbers ranging from 60,000 to 200,000. Laminar separation bubble predictions from Xfoil are also examined.

The low Reynolds $k-\omega$ SST model generated the most accurate aerodynamic performance across the Reynolds number range tested. At the higher end of the investigated range of Reynolds numbers (near 200,000), the SA-BC model predicted the aerodynamic performance the most accurate, especially around the top corner of the drag bucket. At Reynolds numbers of 150,000 and below, the $k_T - k_L - \omega$ and the SA-BC models made poor predictions of the aerodynamic performance for all airfoils tested, with the exception of the NACA0018 airfoil. The $\gamma - Re_\theta$ transition model was accurate at predicting aerodynamic performance for all Reynolds numbers and airfoils tested up until moderate lift coefficients, where over-predictions of drag and under predictions of lift were evident.

The $\gamma - Re_\theta$ transition model was able to predict the flow feature locations within 5% of the experimental results for almost all conditions over the three airfoils tested. At a Reynolds number of 100,000, the $k_T - k_L - \omega$ and the SA-BC models showed much longer laminar separation bubbles and in some cases full separation where it was not evident in the published results. With the increase in pressure gradient or Reynolds number, these two models were able to match a lot closer to the experimental predictions. The Xfoil prediction for separation and transition aligned well with experimental results, although it consistently under predicted the position of reattachment across all airfoils and Reynolds numbers.

Overall, both the low Reynolds $k-\omega$ SST and the $\gamma - Re_\theta$ models can predict aerodynamic characteristics of airfoils at low Reynolds numbers with sufficient accuracy for engineering applications and are recommended in conditions with laminar separation bubbles at Reynolds numbers below 200,000. At a Reynolds number of 200,000, the SA-BC model is also recommended, while the $k_T - k_L - \omega$ model is not recommended for steady RANS simulations at the conditions tested (at and below a Reynolds number of 200,000 and at turbulence intensities below 0.2%).

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